

Randy Connolly, "Fundamentals of Web Development", Pearson Education Robert W Sebesta, "Programming the World Wide Web", Pearson Education Gottfried Vossen, Stephan," Hagemann Unleashing Web 2.0: From Concepts to Creativity", Elsevier Nicholas C Zakas," Professional AJAX", Wrox publications Frank. P. Coyle, "XML, Web Services And The Data Revolution", Pearson Education. James Snell, Doug Tidwell, Pavel Kulchenko, "Programming Web Services with SOAP", O'Reilly publishers. Web Resources: W3schools.com Developer.mozilla.org/en-US/docs/Learn docs.microsoft.com informit.com/articles/ The Relationship Between Web 2.0 and Social Networking https://presiuniv.knimbus.com/user#/home
Topics relevant to "SKILL DEVELOPMENT": Building blog, Social networking or social media sites for Skill Development through Experiential Learning techniques. This is attained through assessment component mentioned in course handout.

Course Code: CSE258	Course Title: Problem Solving Using Python	L-T-P-C	1	0	4	3
	Type of Course: Theory & Integrated Laboratory					
Version No.	1.0					
Course Pre-requisites	Nil					
Anti-requisites	NIL					
Course Description	This course provides the opportunity for the students of Computer Science engineering to develop Python scripts using its powerful programming features like lists, sets, tuples, dictionaries and sets. Students will also be introduced to object oriented programming concepts and packages for data visualization. Topics include: Basics of Python programming, operators and expressions, decision statements, loop control statements, functions, strings, lists, list processing : searching and sorting, nested list, list comprehension, tuples and dictionaries, sets, file handling, exception handling, object oriented programming concepts, modules and packages for data visualization					

Course Objective	The objective of the course is to familiarize the learners with the concepts of Problem Solving Using Python and attain Employability Skills through Experiential Learning techniques.			
Course Out Comes	On successful completion of the course the students shall be able to: Demonstrate problem solving through understanding the basics of python (Application) Manipulate functions and data structures. (Application) Apply Tuple, Dictionaries, File and Exception Handling concepts to solve real time problems (Application) Practice object-oriented programming (Application) Produce data visualization using modules and packages (Application)			
Course Content:				
Module 1	Problem Solving Techniques and Basics of Python Programming	assignments	Quizzes form basics of python	15Sessions
Basics of problem solving techniques, Basics of Python programming, operators and expressions, decision statements, loop control statements.				
Module 2	Function, String and List	Quizzes and assignments	Comprehension based Quizzes and assignments	15 Sessions
Functions, strings, lists, list processing: searching and sorting, nested list, list comprehension				
Module 3	Data Structures, File and Exception handling	Term paper/Assignment	Quizzes form advanced python	15 Sessions
Tuples and dictionaries, sets, file handling, exception handling.				
Module 4	Object-Oriented Programming and Data Visualization	Term paper/Assignment	Application on data visualization	15 Sessions
Object oriented programming concepts, modules and packages for data visualization.				
List of Laboratory Tasks: Each Lab sheets experiments are prepared by level 0 and level 1 module wise.				
Targeted Application & Tools that can be used: Any IDE – PyCharm, VS Code, Python IDE, Spyder, jupyter note book, Google Colab				

Text Book

T1. Ashok Namdev Kamthane and Amit Ashok Kamthane, "Problem Solving and Python Programming", Tata

Mc Graw Hill Edition, 2018.

T2. Charles Dierbach, "Introduction to Computer Science Using Python", Wiley India Edition, 2015.

T3. Reema Thareja, "Python Programming Using Problem Solving Approach", Oxford University Press, 2017.

References

R1. Balagurusamy, "Introduction to Computing and Problem-Solving Using Python", Tata McGraw-Hill, 2016 R2. Y. Daniel Liang, "Introduction to Programming Using Python", Pearson, 2017

E-Resources:

W1. <http://pythontutor.com/>

W2. <https://www.udemy.com/topic/python/>

W3. <https://in.coursera.org/courses?query=python>

W4: <https://puniversity.informaticsglobal.com/login>

Topics relevant to the Employability SKILLS:

problem solving techniques – Function - Object oriented programming - data visualization for Employability Skills through Experiential Learning techniques. This is attained through assessment component mentioned in course handout.